



Faculty of engineering, Cairo University
Electronics and Electrical Communication Department
ELC4046 – Process Control



Lab 3 _ MODIFICATION TO THE TEXTBOOK PID

ELC 4046 | Fall 2026

Submitted to :

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LABORATORY EXERCISE 25

MODIFICATIONS TO THE TEXTBOOK PID

OBJECTIVE: To demonstrate the following modifications:

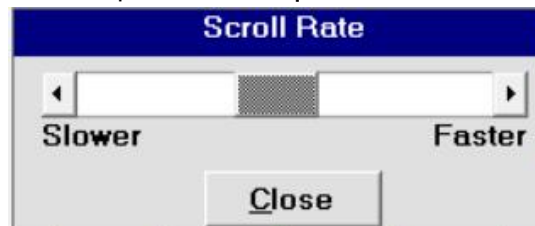
- Derivative on measurement
- Proportional on measurement
- Internal filter
- Setpoint softening
- Interacting vs noninteracting controller
- Bumpless transfer

PREREQUISITE: Completion of Laboratory Exercise 8 PID Controller Characteristics

1. RUNNING THE PROGRAM

Run **PC-ControlLAB**.

Slow down the simulation speed via **View | Scroll Rate** and select a mid-value.

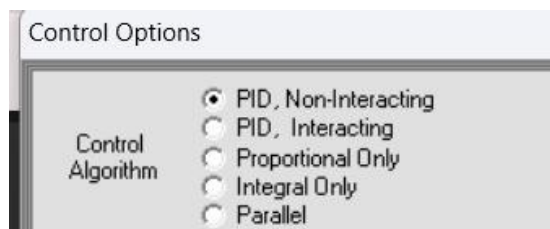


Check the top line of the display. Be sure that you are using the **GENERIC** process model and the **FEEDBACK** control strategy.

Generic Process PC-ControlLAB 3.1 - Evaluation Copy FEEDBACK Control

From the Menu Bar, select **Control | Control Options**.

For Control Algorithm, select **PID, Non-Interacting**.



2. DERIVATIVE ON MEASUREMENT

In **Control | Control Options**, for Deriv on Error or Meas, select **Error**.

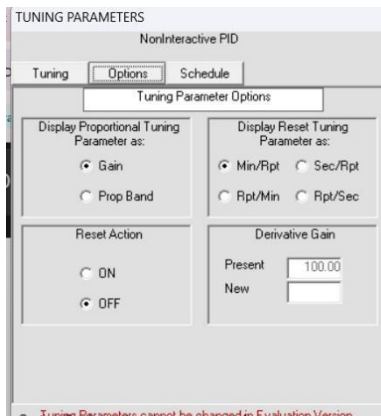


In the **Tune | Options** tab, set the following:

Reset Action	OFF
Derivative Gain	100.0

In the **Tuning** tab, enter the following tuning values:

Gain:	2
Reset, Min/Repeat:	Not relevant (it's OFF)
Deriv, Min:	0.5

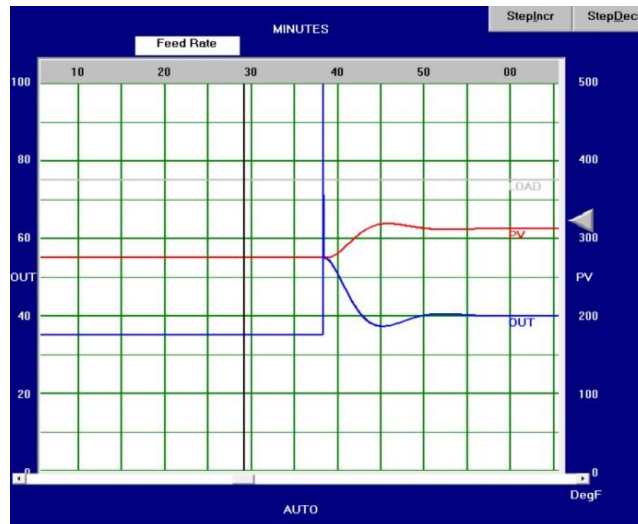


Note that the simulation starts with an initial controller output of 35%. Put the controller in AUTO and change the setpoint to 325.



Observe:

Even with this moderate setpoint change, the controller output “spikes” all the way to 100%, then very quickly drops back to a normal operating range. This is called the *derivative spike*. In a real-world application, you cannot send such a severe shock to the process.

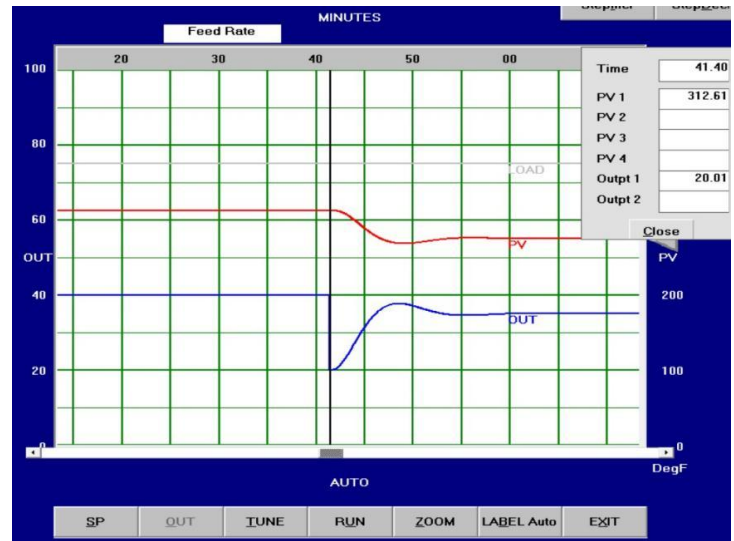
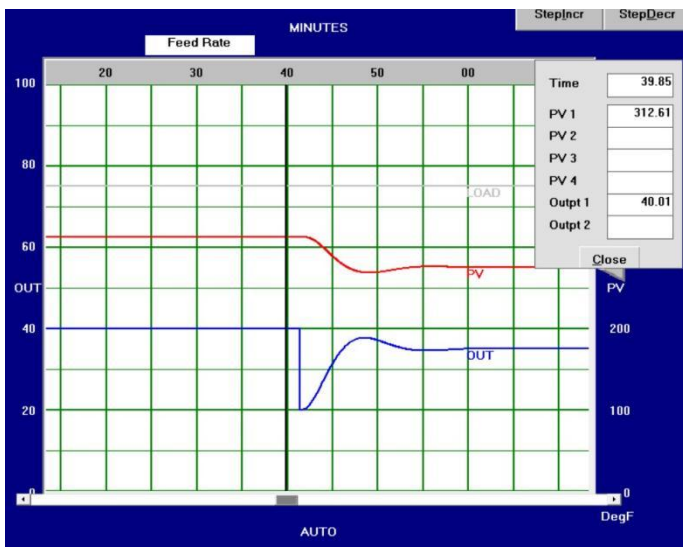


Put the controller in MAN. Select **Control | Control Options**, and select **MEAS** for Deriv on Error or Meas. Put the controller back to AUTO and decrease the setpoint by 50 DegF,

A dialog box titled 'New Set Point' with a text input field containing '275'. Below the input field are two buttons: 'OK' and 'Close'.

Observe:

With derivative on measurement, the immediate response of the controller is the proportional response only – there is no derivative spike.



What is the value of the proportional response?

$$2 * (275 - 325) / 500 * 100 = 20\%$$

3. PROPORTIONAL ON MEASUREMENT (P on PV)

Put the controller in MAN.

In **Control | Control Options**, confirm that, for Prop on Error or Meas, **Error** is selected. This is the default option.

In the **Tune | Options** tab, set the following:

Reset Action ON

In the **Tuning** tab, enter the following tuning values:

Gain: 1.4

Reset, Min/Repeat: 7

Deriv, Min: 0

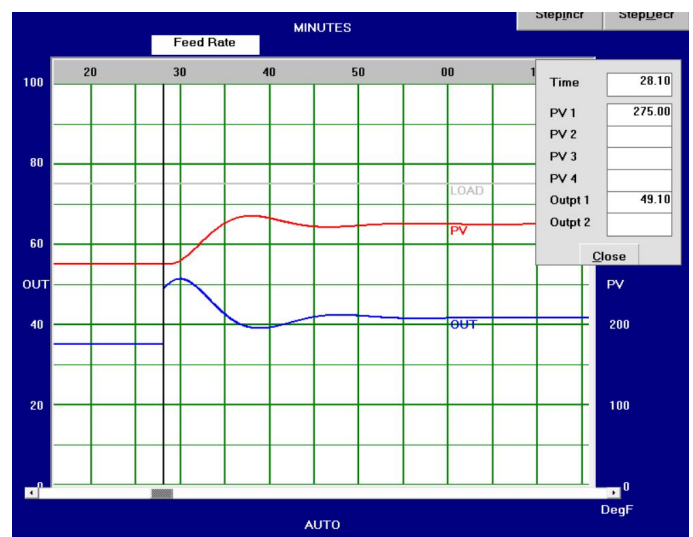
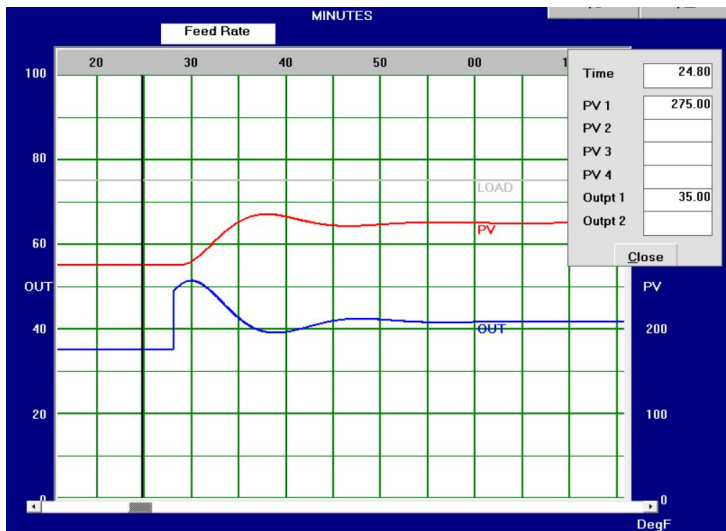
Put the controller in AUTO. Increase or decrease the setpoint by 50 DegF.

New Set Point

New Set Point

What is the immediate change in the controller output?

OUT1= 35 , OUT2=49 , so the change is $OUT2-OUT1= 49\%-35\%=14\%$



This is called the *proportional kick*. Verify that it is equal to the controller gain multiplied by the error (in percentage).

Verification: $P_{kick} = K_c * e = 1.4 * (((325-275)/500)) * 100 = 14\%$

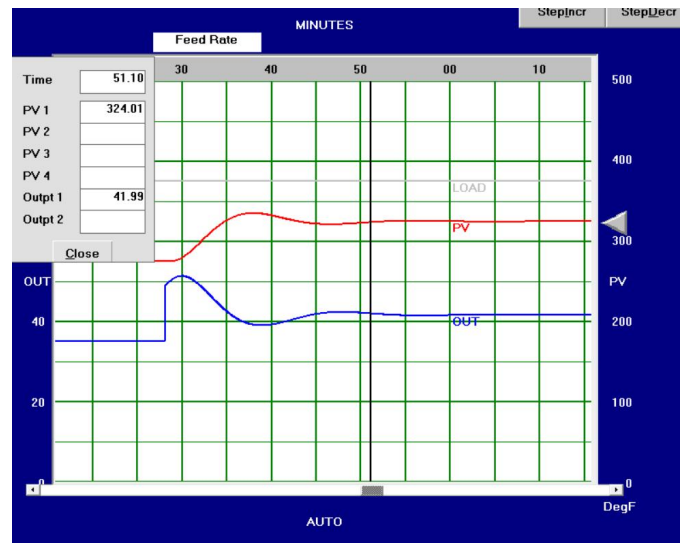
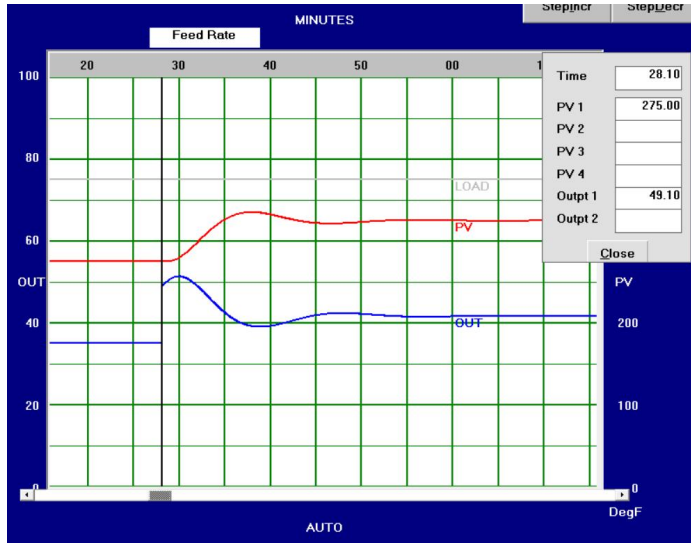
Calculate the ($\pm 2\%$) settling time:

Settling time (T_s) = $98\% * \text{change}$

Change = $(325-275) = 49$

Final PV = $.98 * \text{change} + 275 = 324$

$T_s = 51.1\text{min} - 28.1\text{min} = 23\text{min}$

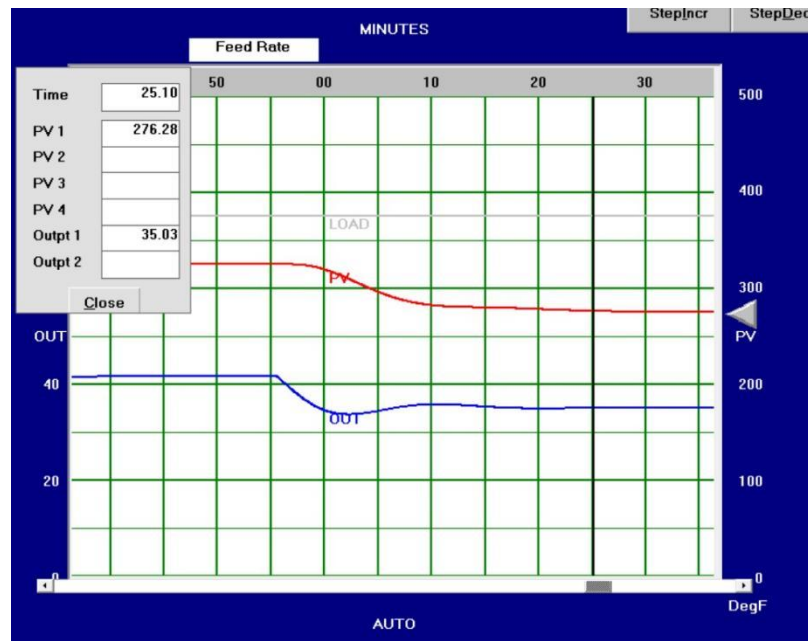


Put the controller in MAN. Select **Control | Control Options**. For Prop on Error or Meas, select **MEAS**. Put the controller back to AUTO and rechange the setpoint by 50 DegF,



Observe:

With proportional on measurement, the immediate response of the controller is a smooth gradual change driven by the integral component. There is no proportional kick. This feature is often favored in the oil and energy industries.



What is the settling time in this case?

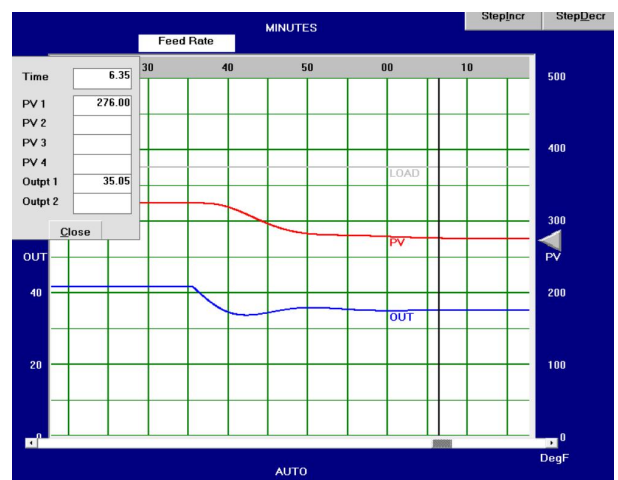
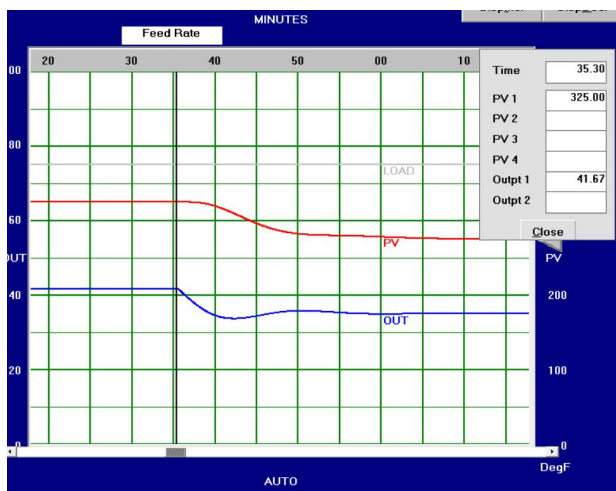
This should confirm that this version of the algorithm is slower, i.e. it does not drive the measurement to the new setpoint as quickly as the proportional on error version would do. This is due to the absence of the initial rapid forcing of the process caused by the proportional kick.

Settling time (T_s) = 98% * change

Change = $(325-275)*0.98 = 49$

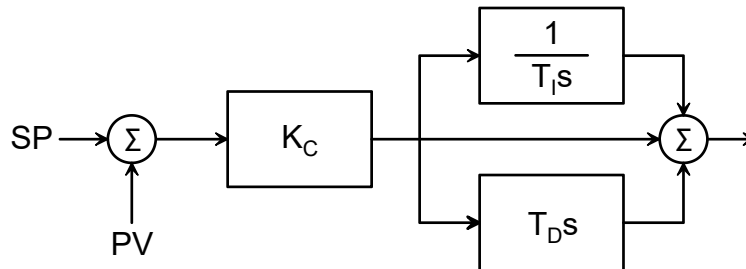
Final PV = $325-98*\text{change} = 276$

$T_s = (60-35.3)\text{min} + 6.35\text{min} = 31.05\text{min}$

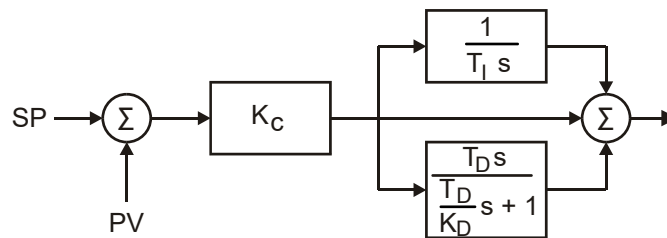


4. INTERNAL FILTER

A block diagram for the textbook PID controller is shown below.



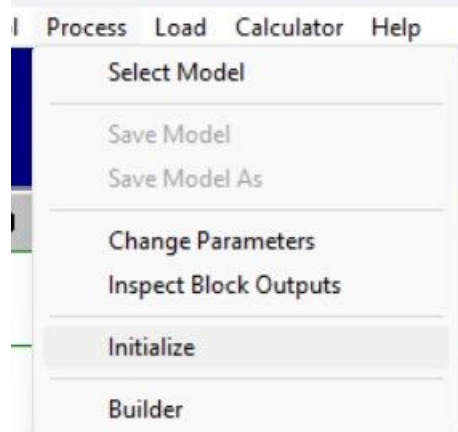
Many commercial manufacturers of controllers, however, place a filter, in the form of a first-order lag, in the derivative portion of the controller. This results in the modified block diagram shown below:



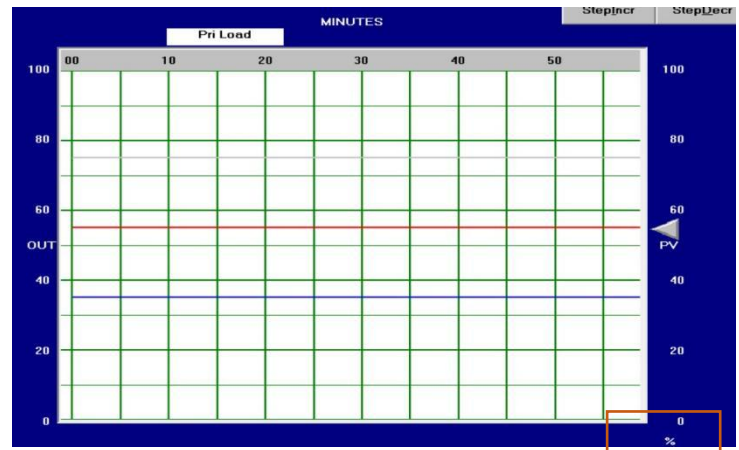
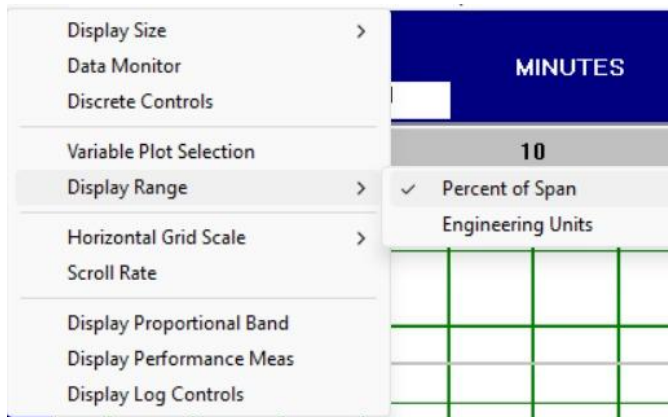
The usual practice is for the time constant of the first order lag to be some fraction of the derivative time. This is accomplished by dividing the derivative time by a factor known as “derivative gain” (K_D). In commercial controllers, the value of K_D is usually in the range of 8 to 15 (with a default value of 10 in most of them).

The purpose of the filter is to reduce the amplification of measurement noise placed on the controller output by the derivative component. With the textbook PID, even a small but sudden change in the measurement will result in a spike in the controller output caused by the derivative unit. With the filter added, the spike height following a change in the measurement will be the value of K_D times the proportional response. The output will then decay to normal with the time constant of the filter.

Select **Process | Initialize** (This returns all options to their original settings.)



Select **View | Display Range | Percent of Span**.



Select **Control | Control Options** and confirm that for Deriv on Error or Meas, **MEAS** is selected.

This is the default option.



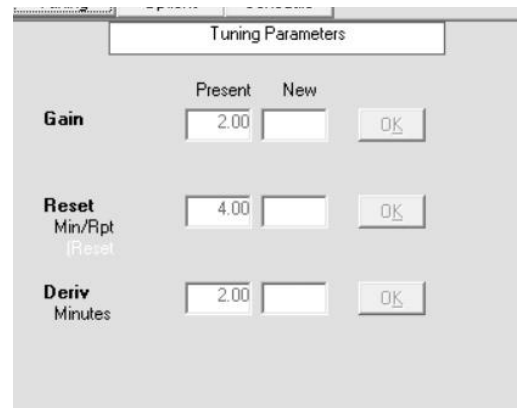
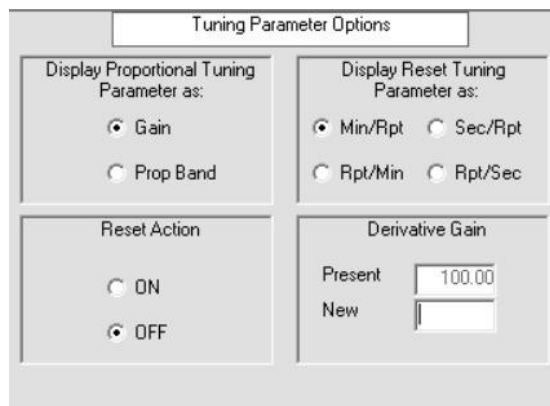
Now eliminate the derivative filtering action by setting the derivative gain to a large value (i.e., small filter time constant).

In the **Tune | Options** tab, set the following:

Reset Action	OFF
Derivative Gain	100.0

In the **Tuning** tab, enter the following tuning values:

Gain:	2
Reset, Min/Repeat:	Not relevant
Deriv, Min:	2



In **Control | Measurement Options**, select **Yes** for “Use substitute value ...?”
Change the controller mode to AUTO.

Feedback PV

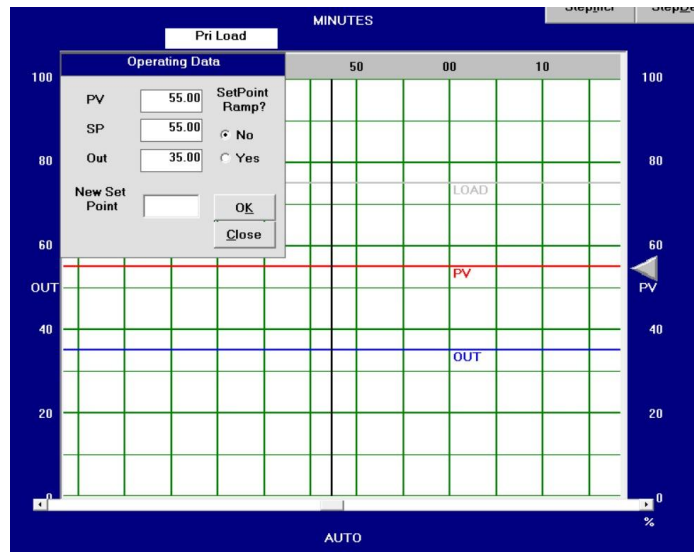
Measurement Filter ☒ No Time Const ☐ Yes 0.00

Use substitute value instead of value from process sensor? ☐ No ☒ Yes Substitute Value 55.00

Use ramped value instead of value from process sensor? ☒ No Ramp Rate ☐ Yes 0.00

Direction EU/min
☐ Down ☒ Stop ☐ Up

Caution: You must confirm each new data value, by pressing Enter, before closing this window.



The controller should be in stable operation with the following values:

Set Point	55
Process Variable	55
Controller Output	35.0

In **Control | Measurement Options**, enter a substitute measurement value of 54.

Feedback PV

Measurement Filter ☒ No Time Const ☐ Yes 0.00

Use substitute value instead of value from process sensor? ☐ No ☒ Yes Substitute Value 54.00

Use ramped value instead of value from process sensor? ☒ No Ramp Rate ☐ Yes 0.00

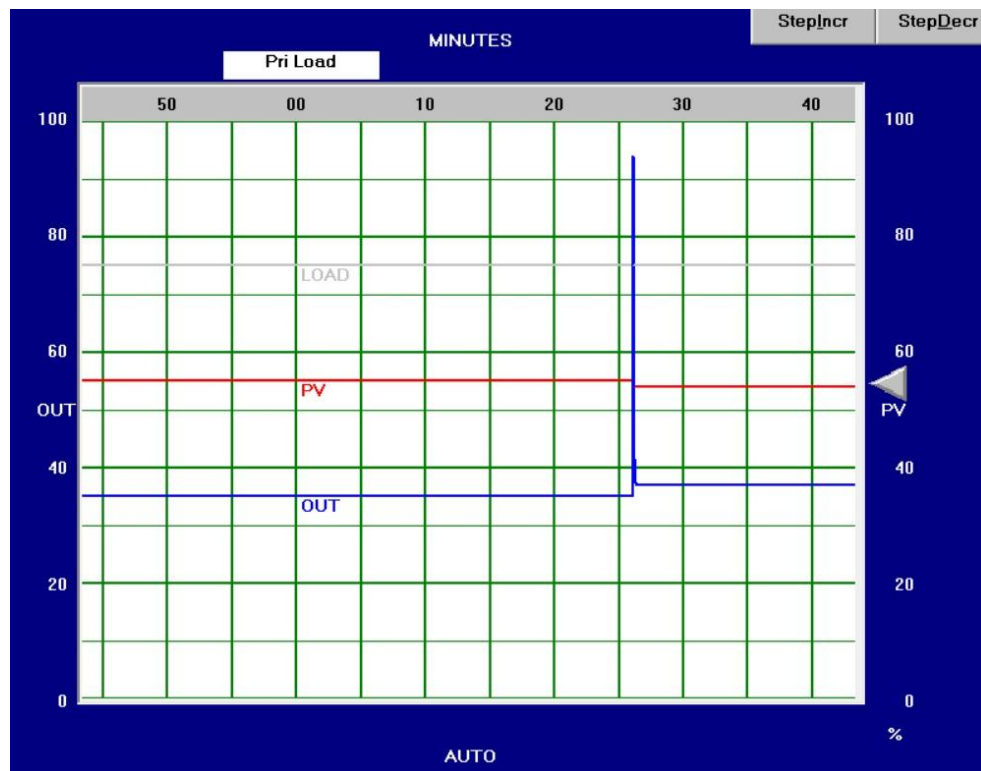
Direction EU/min
☐ Down ☒ Stop ☐ Up

Caution: You must confirm each new data value, by pressing Enter, before closing this window.

Close

Observe:

Even with this small process variable change, the controller output experiences an excessive derivative spike, then very quickly drops back to a normal control range. In a real world application, this is completely unacceptable.



Through the **Tune | Options** tab, set:

Derivative Gain 10.0

Tuning Parameter Options

Display Proportional Tuning Parameter as:

☒ Gain

☐ Prop Band

Display Reset Tuning Parameter as:

☒ Min/Rpt ☐ Sec/Rpt

☐ Rpt/Min ☐ Rpt/Sec

Reset Action

☐ ON

☒ OFF

Derivative Gain

Present: 10.00

New:

In **Control | Measurement Options**, enter a substitute measurement value of 53.

Measurement Options

Feedback PV

Measurement Filter ☒ No ☐ Yes Time Const 0.00

Use substitute value instead of value from process sensor? ☒ No ☐ Yes Substitute Value 53

Use ramped value instead of value from process sensor? ☒ No ☐ Yes Ramp Rate 0.00 EU/min

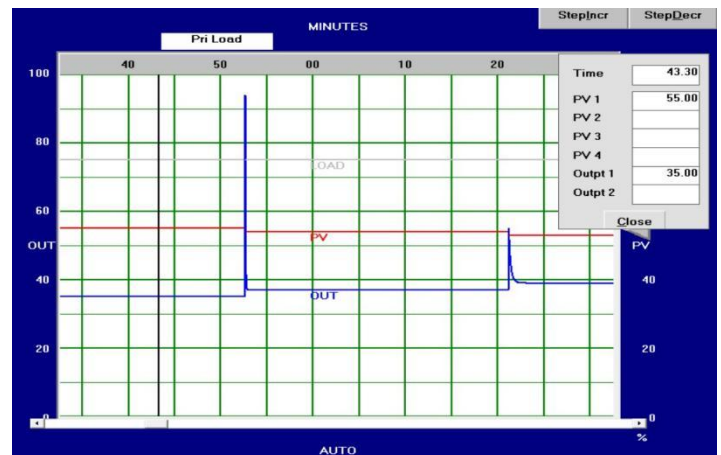
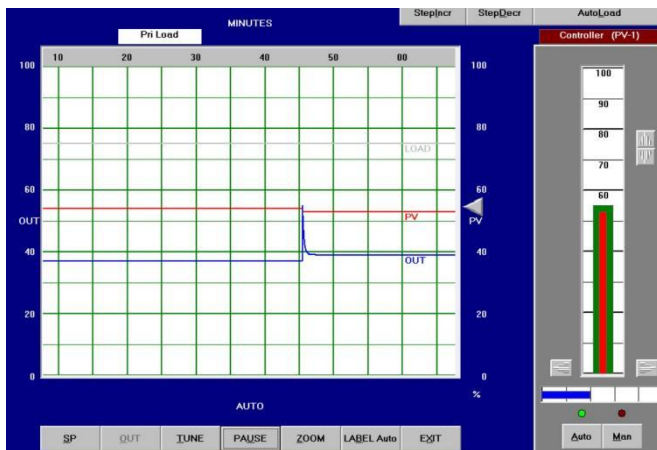
Direction ☒ Down ☐ Stop ☐ Up

Caution: You must confirm each new data value, by pressing Enter, before closing this window.

Close

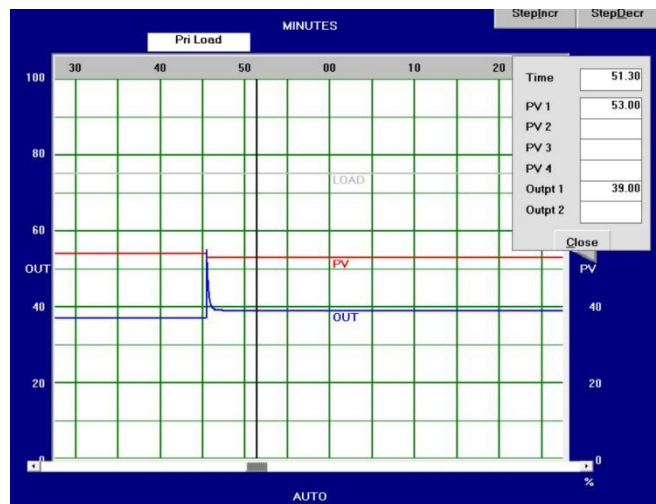
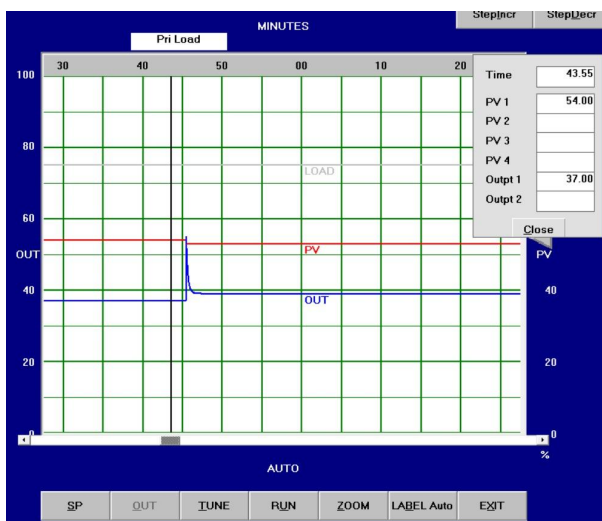
Observe:

This time the controller output spike is limited.



Calculate the proportional kick:

From the out value the Pkick: $(39-37)=2\%$



The now reasonable derivative spike should be:

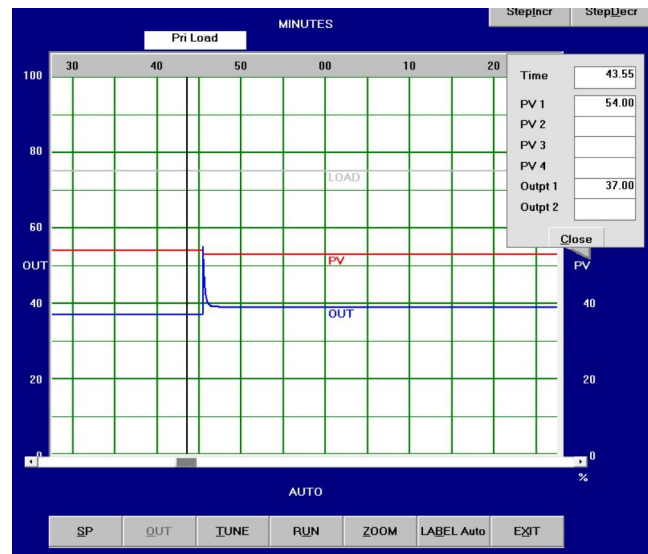
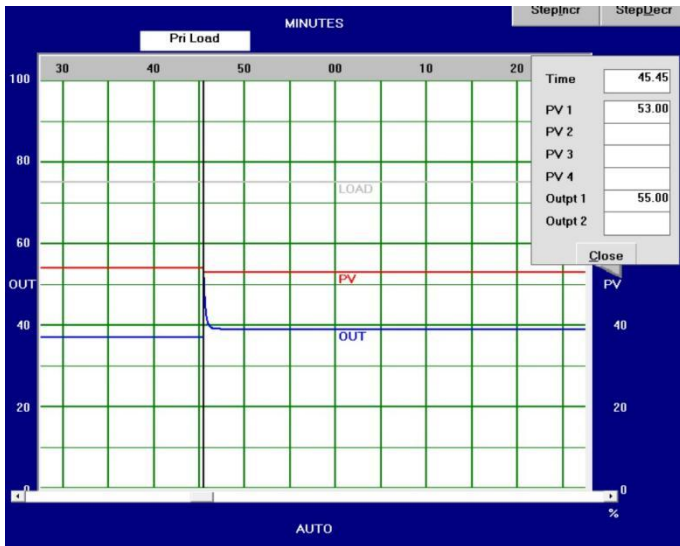
$$\text{PV change (\%)} \times \text{Controller Gain} \times \text{Deriv Gain} \\ 1 \times 2.0 \times 10 = 20\%$$

Measure the actual immediate change in the controller output:

$$\text{Immediate change} = 55 - 37 = 18\%$$

=

$$\text{derivative spike} + \text{proportional kick} = 16\% + 2\% = 18\%$$



The immediate change in controller output should equal:
the proportional kick + the derivative spike

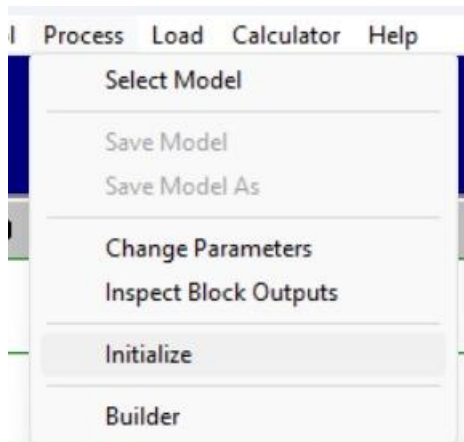
(Quantization error in the digital algorithm may cause some deviation between the actual value and the calculated value).

Also note that the return to the normal operating range is not as abrupt, due to the filter.

5. SETPOINT SOFTENING

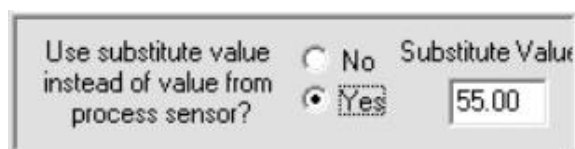
5.1 Setpoint Softening Demonstration

Select **Process** | **Initialize**.

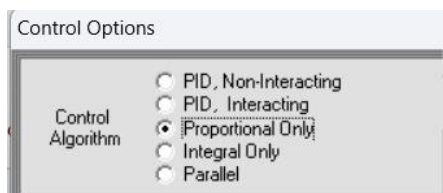


Select **Control** | **Measurement Options**.

Select **Yes** for “Use substitute value instead of value from process sensor?”

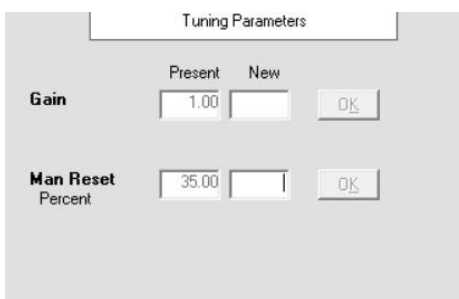


From **Control** | **Control Options**, select **Proportional Only** control algorithm.

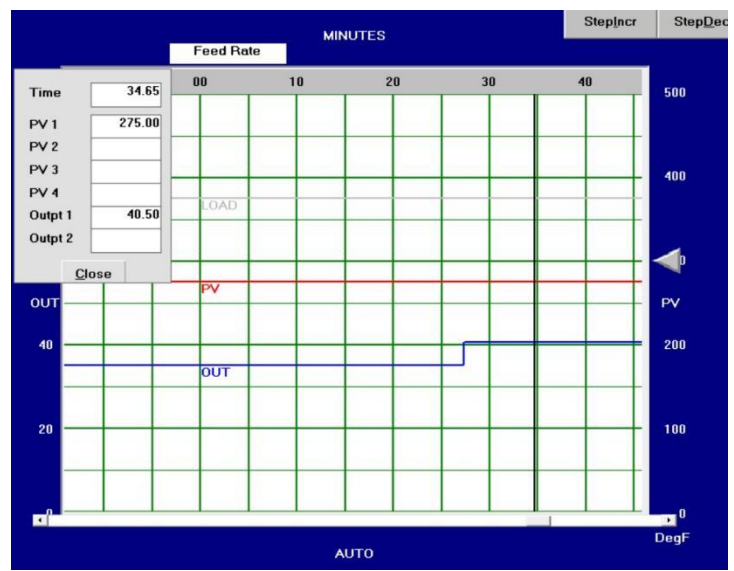
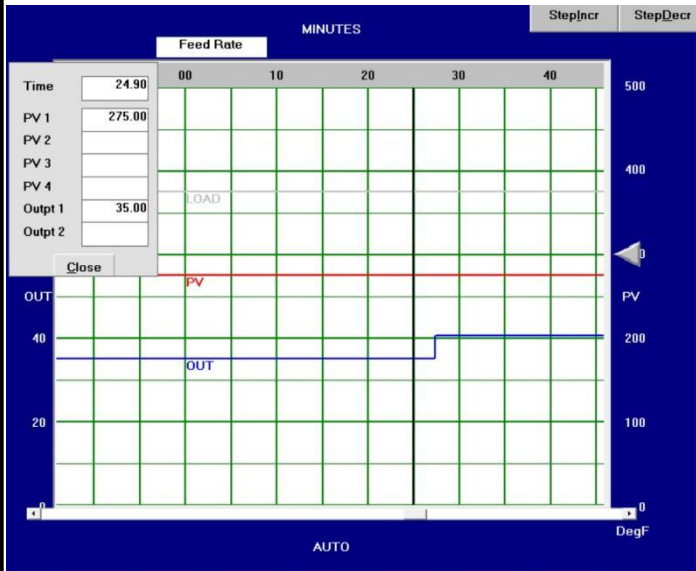


In the **Tuning** tab, set:

Gain = 1



What would be the change in the controller output (proportional kick), if the setpoint is abruptly increased by 10%?



Put the controller in AUTO. Increase the setpoint to 325 DegF, enabling the **SetPoint Ramp** option (enter a rate of 5 DegF/min) before making the setpoint change. We use setpoint ramping as the more common setpoint filtering is not supported by this version of the simulation software.

New Set Point

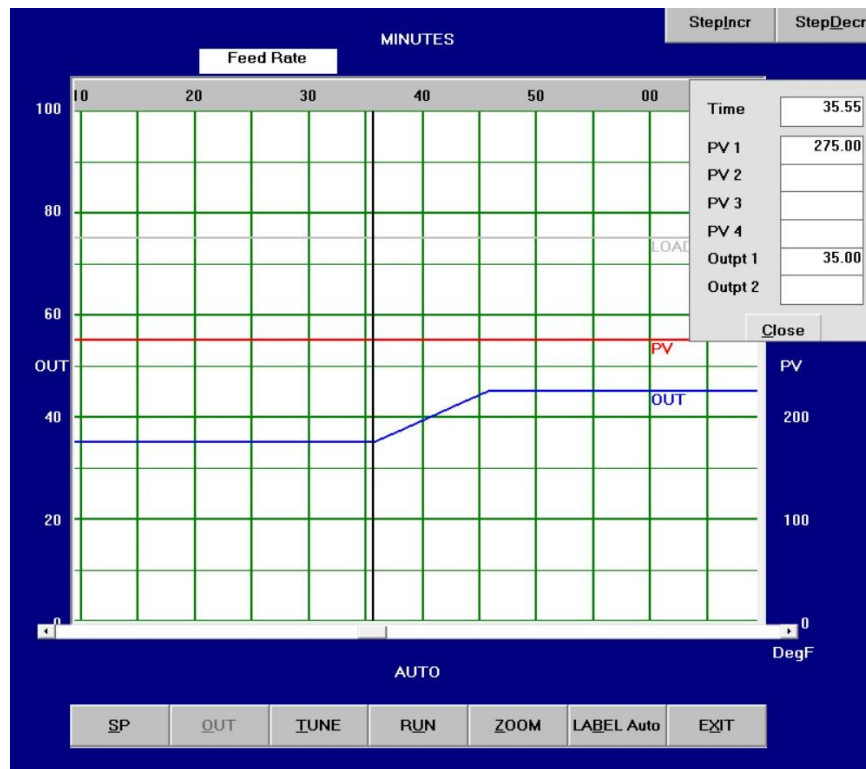
325

Ramp Rate: EU/min

5

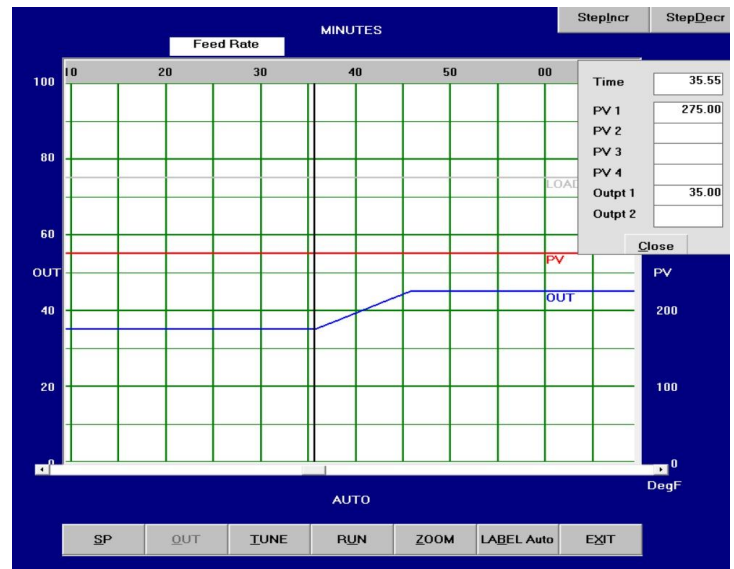
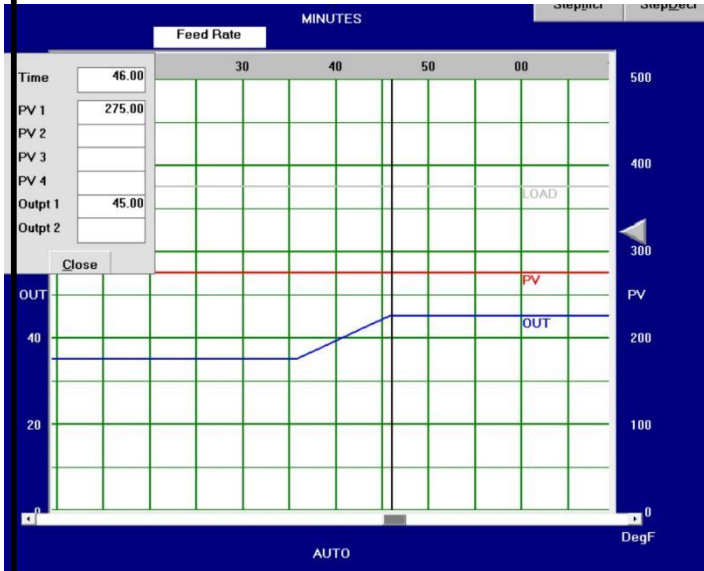
Observe:

The effective setpoint used by the controller ramps up towards the new value. The controller output follows the setpoint waveform.



How long does it take for the controller to reach its new steady value?

$$46.00\text{min} - 35.55\text{min} = 10.45\text{min}$$



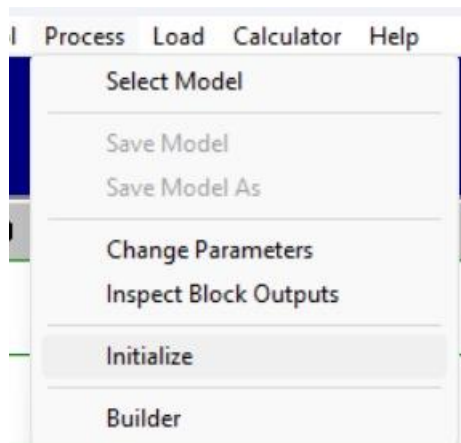
Does this match your expectation?

Yes, Since the setpoint increases gradually from 275 to 325 at a constant rate of 5 Degrees/min, the total ramp duration is determined by dividing the total SP increase by the ramp rate. The required SP change is 50 Degree, therefore the ramp takes 10 minutes to complete

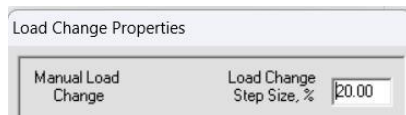
Note that in most commercial controllers, the effective setpoint changes are transparent to the user and are calculated behind the scenes.

5.2 Setpoint Softening, A Second Example

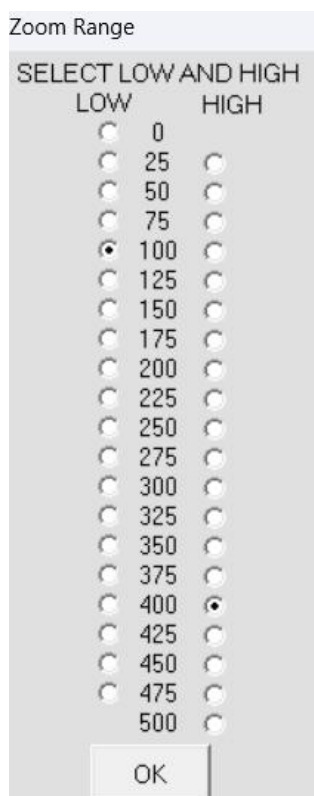
Select **Process** | **Initialize**.



Select **Load** and set Manual Load Change Step size to 20%.



Press **ZOOM** and select 100 and 400 respectively.



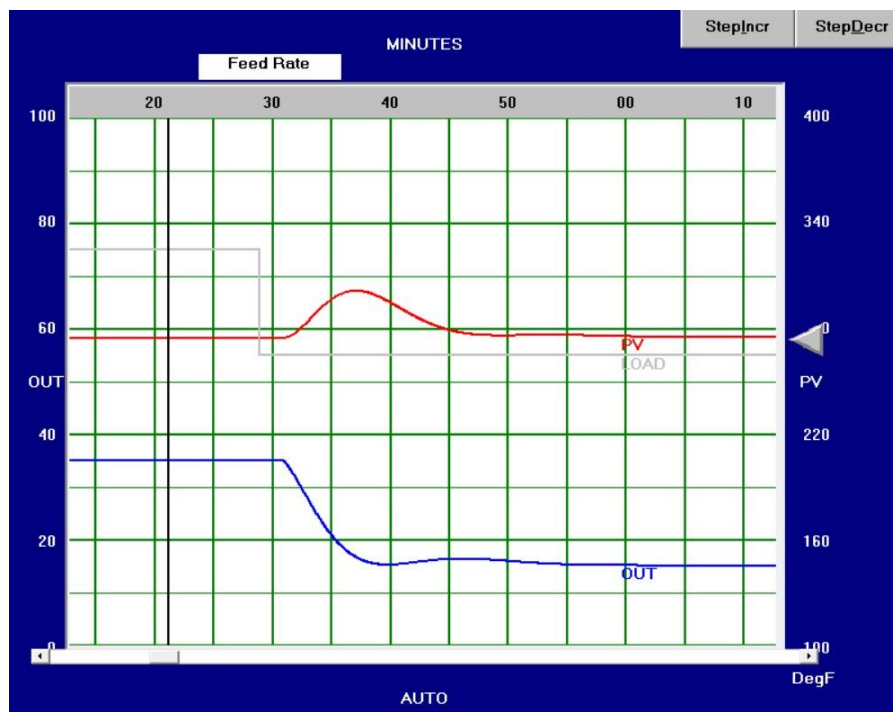
In the **Tuning** tab, enter the following tuning values:

Gain: 2
Reset, Min/Repeat: 6
Deriv, Min: 0

The screenshot shows a 'Tuning Parameters' dialog box with three rows of controls. Each row has a label, a 'Present' value field, a 'New' value field, and an 'OK' button. The first row is for 'Gain', with 'Present' at 2.00 and 'New' empty. The second row is for 'Reset Min/Rpt', with 'Present' at 5.00 and 'New' empty. The third row is for 'Deriv Minutes', with 'Present' at 1.00 and 'New' empty.

	Present	New	
Gain	2.00		OK
Reset Min/Rpt	5.00		OK
Deriv Minutes	1.00		OK

Put the controller in AUTO. Make a 20% load change. Press **StepDecr** once.

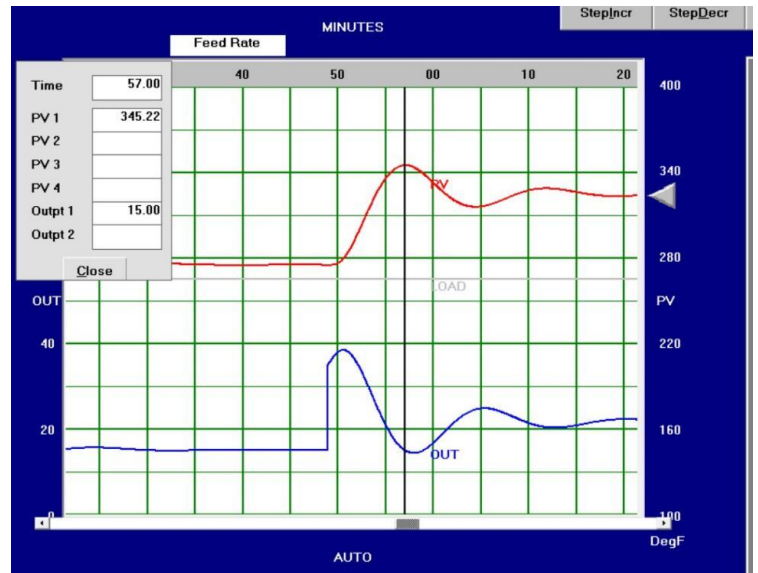


Observe:

A good response to a load upset.

Test the controller by making a setpoint change. Change the setpoint to 325 DegF.

New Set Point



Calculate the process variable overshoot:

$$345.22 - 275 - (325 - 275) / (325 - 275) * 100 = 40.44\%$$

(For a damped response, the industry standard is that a well-tuned controller will bring the process variable to the setpoint in at most one cycle and with an overshoot not exceeding 5%)

Bring the setpoint back to 275 DegF. Wait till the response settles.

New Set Point

Once more, increase the setpoint to 325 DegF, enabling the **SetPoint Ramp** option this time (enter a rate of ~4 DegF/min) before making the setpoint change.

New Set Point

Operating Data

PV SetPoint Ramp? ☐ No ☒ Yes

SP

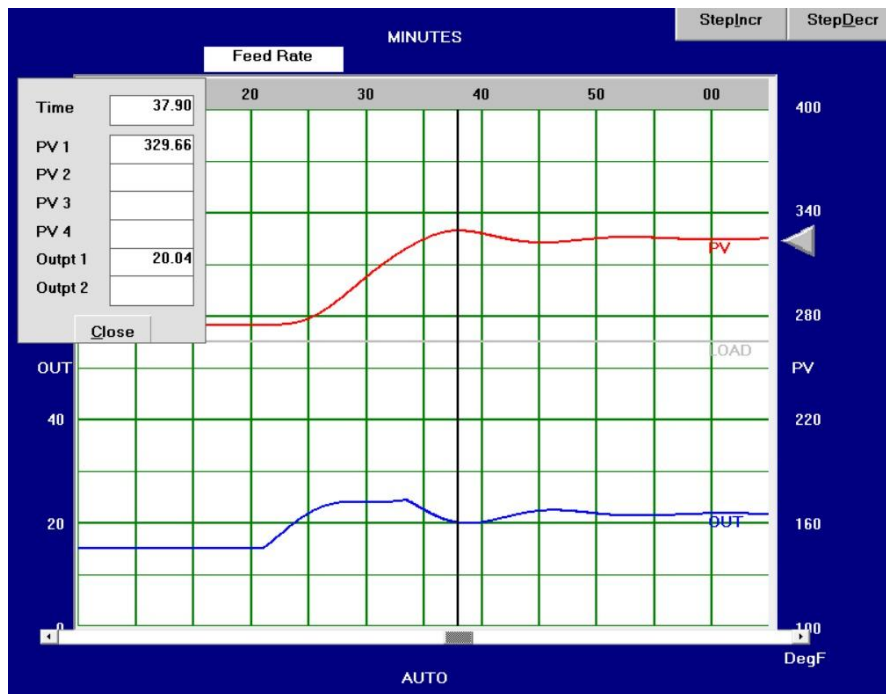
Out

New Set Point

Ramp Rate: EU/min

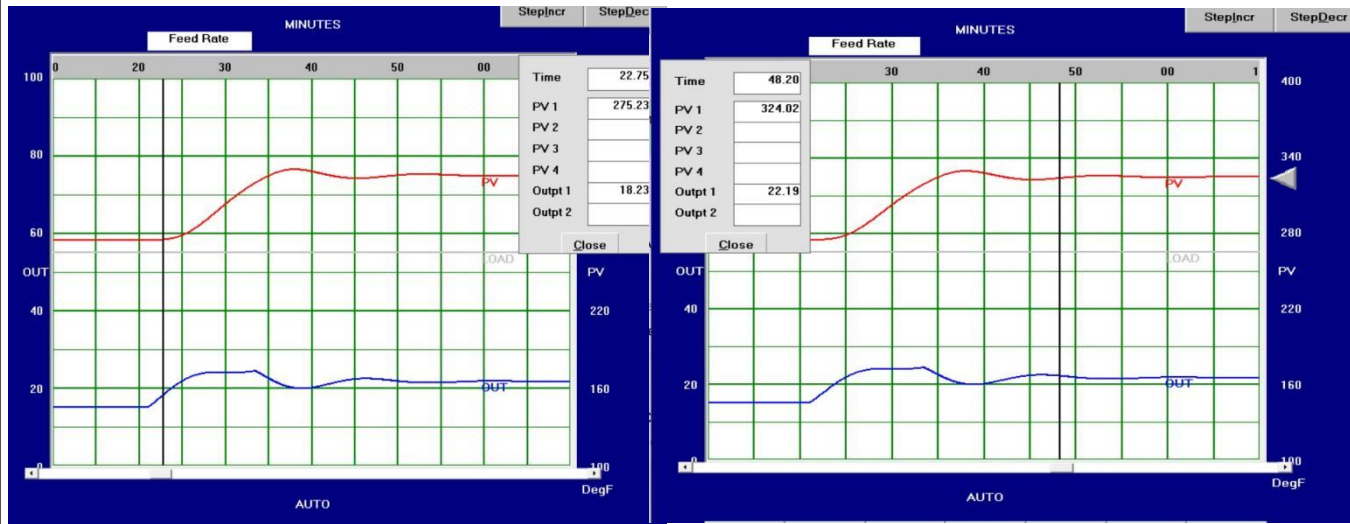
Calculate the process variable overshoot with setpoint ramping:

$$329.66 - 275 - (325 - 275) / (325 - 275) * 100 = 9.32\%$$



Also note that the settling time is shorter in this case.

$$48.20\text{min} - 22.75\text{min} = 25.45\text{min}$$



What you should have observed:

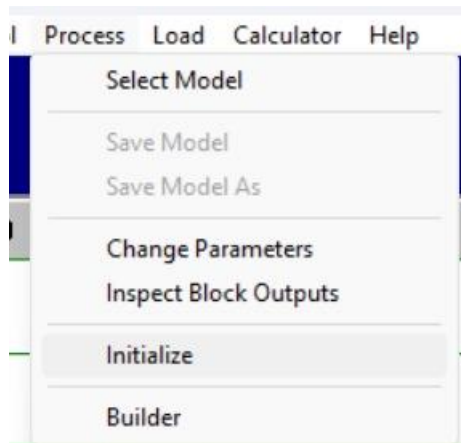
A controller tuning that provides excellent load upset rejection can produce large overshoot for setpoint changes. With setpoint softening, when a step setpoint change is made towards a new target value, the effective setpoint used by the controller is changed rather gradually (ramped) from its present value to the target value at a specified rate. This will result in a more gradual change in the process variable, considerably reducing overshoot.

Therefore, it is practical to tune the controller for satisfactory load upset response and utilize setpoint softening in order not to sacrifice setpoint tracking. This is called two degrees of freedom controller.

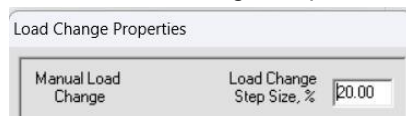
Note that this advantage comes with the complexity of adding one more additional tuning parameter to the algorithm: the setpoint ramp rate, or the setpoint lead-lag ratio for the more common setpoint lead-lag filter.

6. INTERACTING VS NONINTERACTING CONTROLLER

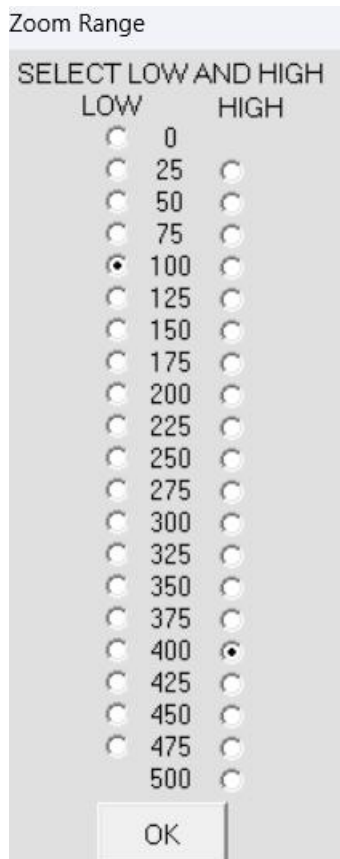
Select **Process** | **Initialize**.



Select **Load** and set Manual Load Change Step size to 20%.



Press **ZOOM** and select 100 and 400 respectively.



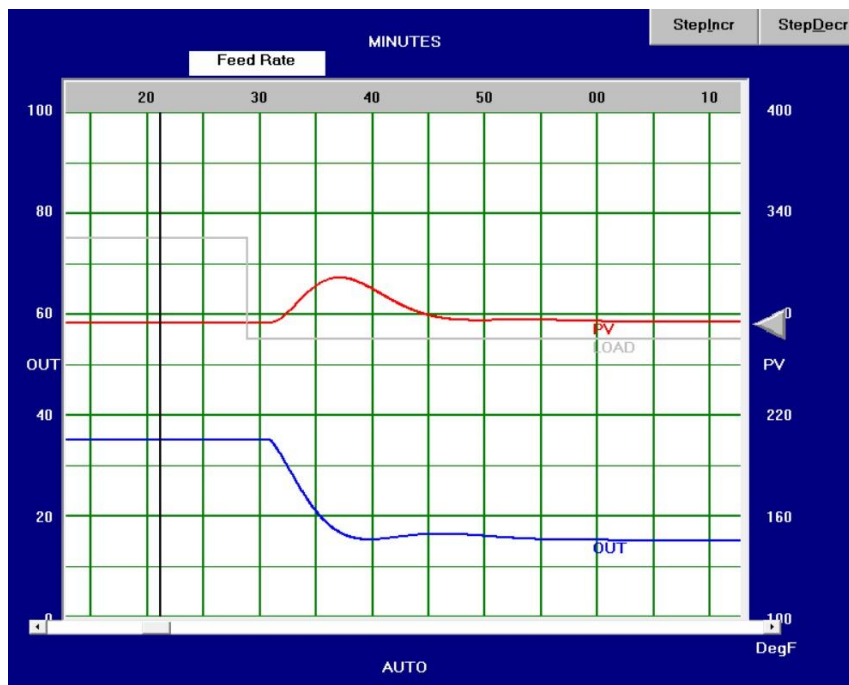
In the **Tuning** tab, enter the following tuning values:

Gain:	2
Reset, Min/Repeat:	5
Deriv, Min:	1

The screenshot shows a 'Tuning Parameters' dialog box with three rows of controls. Each row has a label, a 'Present' value field, a 'New' value field, and an 'OK' button. The first row is for 'Gain', with 'Present' at 2.00 and 'New' empty. The second row is for 'Reset Min/Rpt', with 'Present' at 5.00 and 'New' empty. The third row is for 'Deriv Minutes', with 'Present' at 1.00 and 'New' empty.

Parameter	Present	New	Action
Gain	2.00		OK
Reset Min/Rpt	5.00		OK
Deriv Minutes	1.00		OK

Put the controller in AUTO. Make a 20% load change by pressing **StepDecr** once. Take a screenshot of the response and save it to another file. Return the load to its original value.



Calculate the apparent (entry) tuning values for the interacting controller.

$$\lambda = \frac{I}{2} + \sqrt{\frac{I}{4} - \frac{T_D}{T_I}}$$

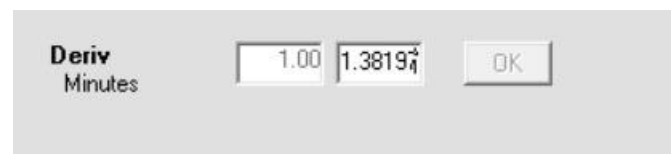
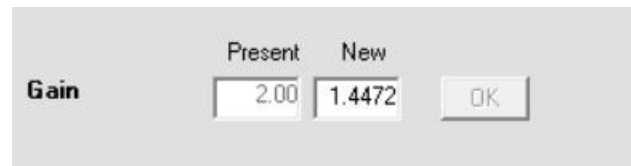
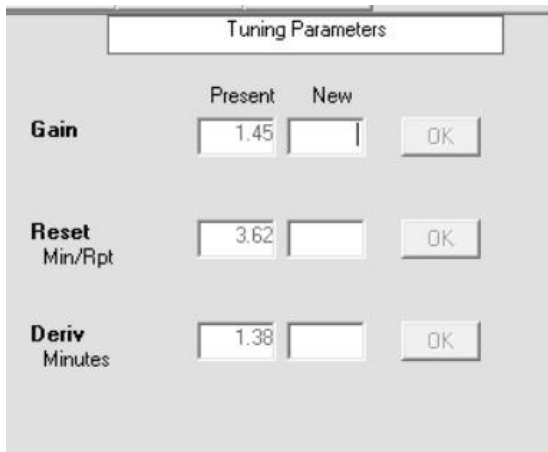
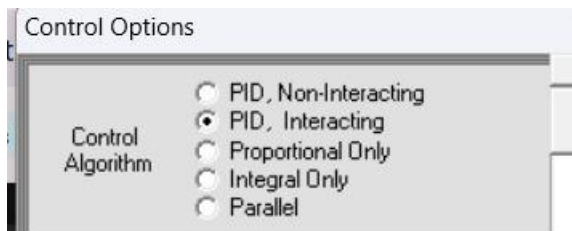
$$\lambda = 0.7236$$

$$\hat{K}_C = \lambda * K_C = 1.4472$$

$$\hat{T}_I = \lambda * T_I = 3.618$$

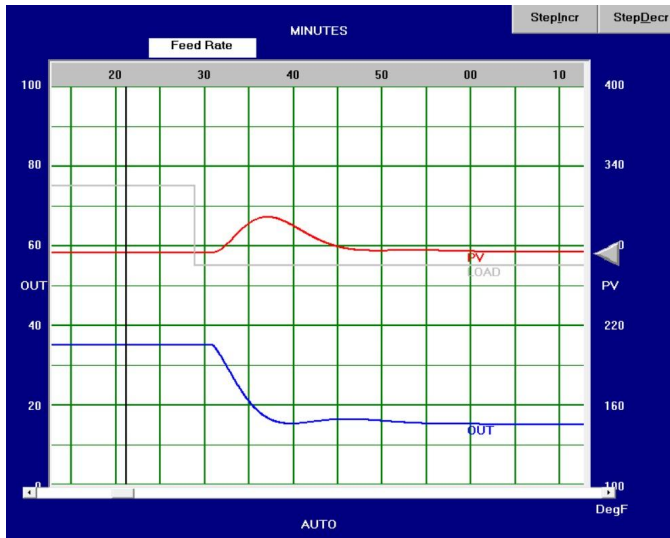
$$\hat{T}_D = T_D / \lambda = 1.38197$$

Wait till PV settles. Switch the controller to MAN. From **Control | Control Options**, select **PID, Interacting** control algorithm. In the **Tuning** tab, enter the values calculated above. Put the controller in AUTO.



Make the exact same load change above. Compare with the saved screenshot. The two responses should essentially be the same.

The two responses show nearly identical dynamic behavior, with comparable overshoot, settling characteristics, and final steady-state values



7. BUMPLESS TRANSFER

7.1 With Setpoint Tracking

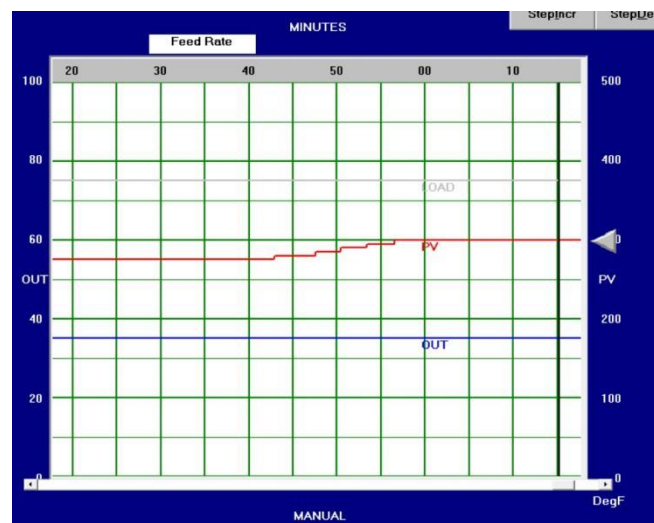
Select **Process | Initialize**.

From **Control | Control Options**, select **YES** for Set Point Tracking.

Set Point Tracking ☐ No ☒ Yes

With the controller in MAN, go to **Control | Measurement Options**, select **Yes** for “Use substitute value ...?”, then successively enter the following measurement values: 280, 285, 290, 295, 300.

Notice how the setpoint tracks the process variable. Press **Close** to remove the Measurement Option box.



Try changing the setpoint via the **SP** button or by moving the arrow to the right of the display. As the setpoint is now tracking the measurement (as long as the controller is in MAN), this should not be possible. Switch the controller to AUTO.

The setpoint cannot be modified while the controller is operating in MAN mode with setpoint tracking enabled

Observe:

Since –with setpoint tracking– there is always no error when we switch from MAN to AUTO, the controller output does not change at all. You can now change the setpoint, and the controller will act accordingly.

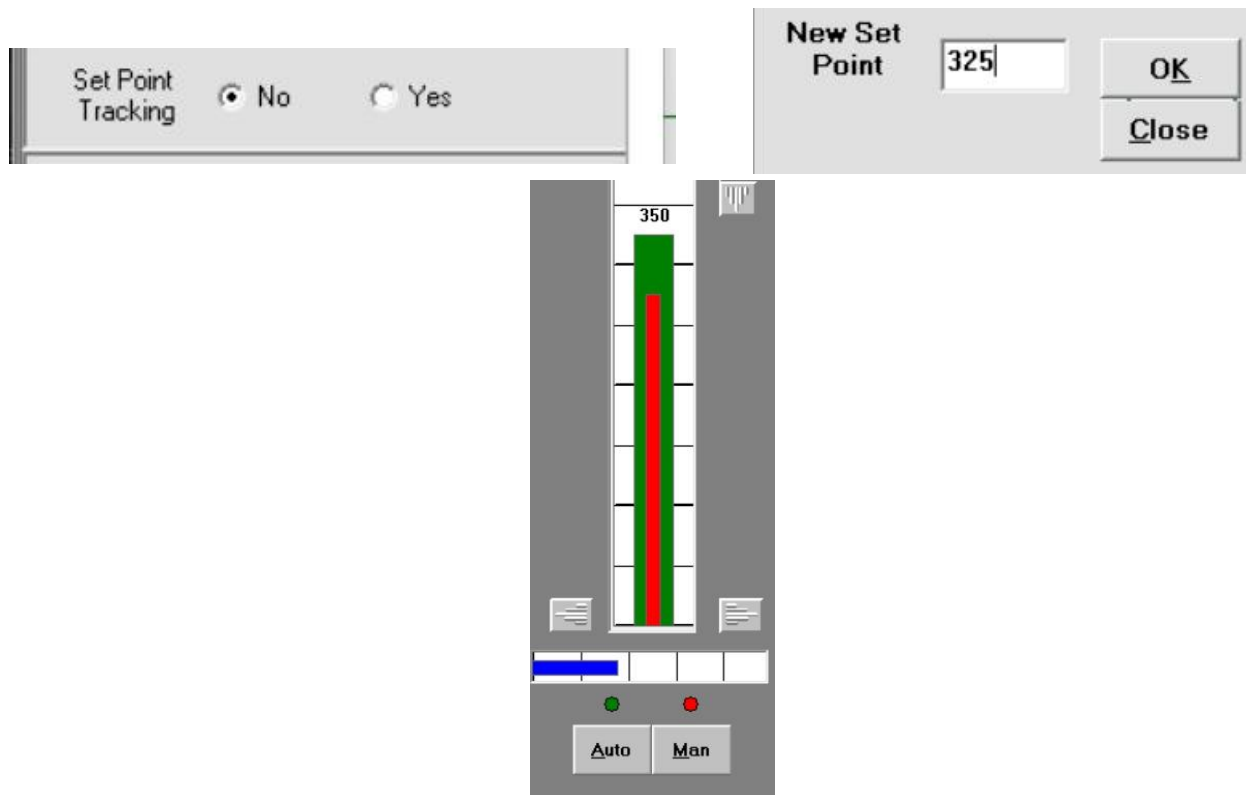
7.2 Without Setpoint Tracking

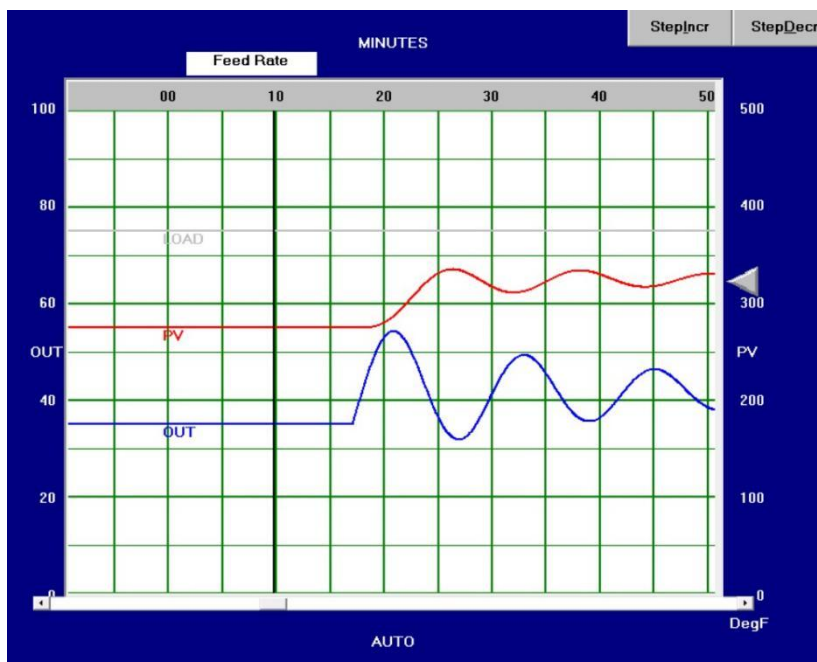
Select **Process | Initialize**.

Select **Control | Control Options** and confirm that for Set Point Tracking, **NO** is selected.

With the controller in MAN, increase the setpoint to 325. Notice that you can now manipulate the set point since setpoint tracking is not enabled. You have just introduced an error of 10%.

Switch the controller to AUTO.





Does the controller output exhibit a bump due to the proportional kick?

The absence of a sudden change in controller output confirms that no proportional kick occurs

This demonstrates the MAN/AUTO bumpless transfer feature that all commercial controllers now have.